Thermal Simulation Report

Siliconware Precision Industries Co., Ltd

SPIL Internal Use

16x16 TFBGA288

(Device name: A9123LCT2A-A0D, 2L Sub, 4L PCB)

RTS13041, Mar. 01, 2013

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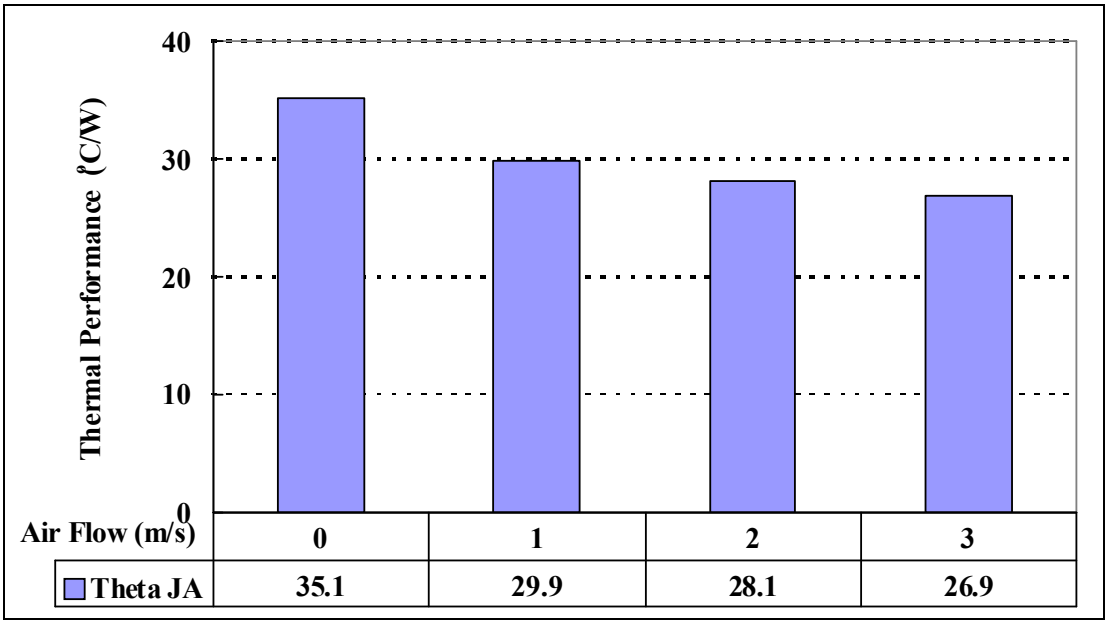
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1. Introduction

In this report, we present the thermal characteristic for 288-pin TFBGA package mounted on 2-layers PCB using the Finite Element Modeling (FEM) method. The junction-to-ambient thermal resistance is utilized to evaluate thermal performance for this package and then to predict the power dissipation capability by given ambient and maximum junction temperatures.

Result Summaries



Thermal Performance of TFBGA Package under Forced Convection



Thermal Resistance of TFBGA under Natural Convection

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3. Conclusion

From the simulation results of temperature contours, we can obtain the chip junction temperatures (T_J) and the case temperatures (T_C). Since the input power (P) and ambient temperature (T_A) are controlled variables, we can calculate the shown above thermal resistance of θ_{JA} according to $\theta_{JA} = \frac{T_J - T_A}{P}$ and assuming ambient temperature of 25°C and maximum junction temperature of 125°C, we can predict the power dissipation capability of this package as below table.

Air Flow (m/s)	0	1	2	3
Power Dissipation (watt)	2.85	3.34	3.56	3.72

4. Assembly Descriptions

Package	Type	TFBGA288
	Device Name	A9123LCT2A-A0D
	Substrate Drawing No.	I-E0959 (2L) Rev.A
	Dimension (L x W)	16x16 mm
	Thickness	1.3 mm (include ball)
	Chip Size	6.274 x 5.2 x 0.119 mm (247 x 205 x 4.7 mil)
	Die Attached Thickness	0.0254 mm
	Ball Pitch	0.8 mm
	Number of Cu Layer-Substrate	2 layers
PCB	PCB Dimension (L x W)	4” x 4.5”
	PCB Thickness	1.6 mm
	Number of Cu Layer-PCB	4 layers (2S2P)

5. Thermal Modeling Method

Thermal resistance represents the capability of an IC package to carry out the heat inside an IC chip. It is a complex function of package structure, material property, input power, environment variables such as air flow speed and PCB layers that can be analyzed by finite element modeling (FEM) techniques. In this simulation a commercial Finite Element Analysis (FEA) program, ANSYS, was used to create the 3D thermal model stacked by brick elements. Considering the package structure symmetry, only the quarter package structure was modeled to save CPU calculating time. All the materials are assumed isotropic with constant physical property and all material interfaces are also assumed perfect joined. The heat flux is applied on the top surface of the die to simulate heating power and constant convection coefficients applied on the exterior surface of this model. The physical properties and simulation conditions are summarized as below.

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A. Simulation Conditions

Input Power	2W
Test Board (PCB)	4 Layers
Control Condition	Air Flow = 0, 1, 2, 3 m/s

B. Material Property

Item		Material	Thermal Conductivity K (W/m-k)
Package	Die	Si	147
	Silver Paste	FH-900NT	0.2
	Mold Compound	KE-G2250	0.9
Solder Ball		SAC	55
Substrate		Cu	400
		BT	0.2
PCB		Cu	400
		FR4	0.2

Appendix 1-Terminology

The major thermal dissipation paths can be illustrated as following

T_J: the maximum junction temperature;

T_A: the ambient or environment temperature;

T_C: the maximum compound surface temperature;

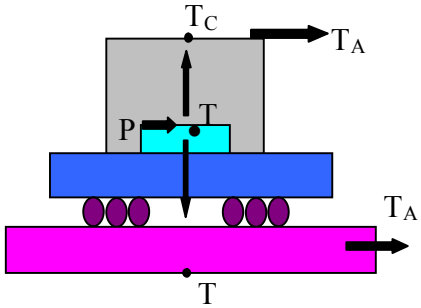
T_B: the maximum surface temperature of PCB bottom;

P: total input power

The thermal parameter can be define as following

1. *Junction to ambient thermal resistance,* θ_{JA}**, defined as:**

$$\theta_{JA} = \frac{T_J - T_A}{P} \ ;$$

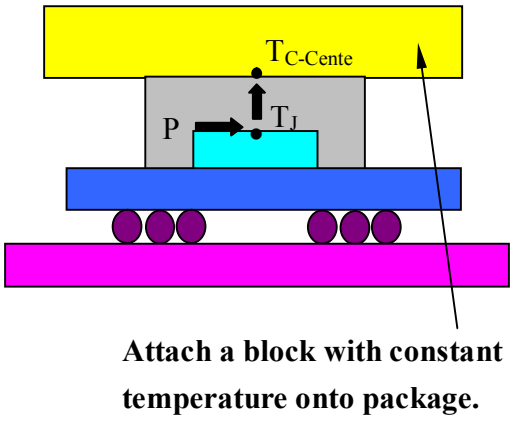


Thermal Dissipation of TFBGA Package

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2. *Junction to case thermal resistance,* θ_{JC}**, defined as:**

$$\theta_{JC} = \frac{T_J - T_C}{P} \ ;$$



Attach a block with constant temperature onto package.

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